# Customer Behaviour Analysis

Retail / E-commerce dataset — customer transaction level data for behavior analysis & business decision-making.

### Numerical

Age, Purchase Amount, Previous Purchases

### Categorical

Category, Payment Method, Season

### Level

Customer Transaction Level


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# Why This Data Matters

### Business Goals

- → Identify high-value customers
- → Understand purchase patterns
- → Optimize discount strategies
- → Increase Customer Lifetime Value (CLV)

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# The Business Problem

How can a retail company understand purchasing behavior to **increase revenue, improve retention, and optimize marketing?**

### Low Retention

Who are frequent vs. one-time buyers?

### Ineffective Campaigns

Do discounts actually drive purchases?

### Seasonal Demand

Which season drives maximum sales?

### Segmentation

High-value, frequent, discount-sensitive customers

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# Data Dictionary

```
<class 'pandas.DataFrame'>
RangeIndex: 5050 entries, 0 to 5049
Data columns (total 17 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Customer ID           5050 non-null   int64
 1   Age                   5050 non-null   int64
 2   Gender                5050 non-null   str
 3   Item Purchased        5050 non-null   str
 4   Category              5050 non-null   str
 5   Purchase Amount (USD)  4494 non-null   float64
 6   Location              5050 non-null   str
 7   Size                  4680 non-null   str
 8   Color                 5050 non-null   str
 9   Season                5050 non-null   str
 10  Review Rating         4449 non-null   float64
 11  Subscription Status   5050 non-null   str
 12  Shipping Type         5050 non-null   str
 13  Discount Applied      5050 non-null   str
 14  Previous Purchases    4502 non-null   float64
 15  Payment Method        5050 non-null   str
 16  Frequency of Purchases  5050 non-null   str
dtypes: float64(3), int64(2), str(12)
memory usage: 670.8 KB
```

18 columns covering customer demographics, product details, and transaction behavior.

| Column | Description |
| --- | --- |
| Customer ID | Unique identifier |
| Age / Gender | Customer demographics |
| Category | Clothing, Electronics, etc. |
| Purchase Amount | Amount spent (USD) |
| Season | Winter, Summer, Spring, Fall |
| Discount Applied | Yes / No |
| Previous Purchases | Number of past purchases |
| Subscription Status | Yes / No |
| Payment Method | Cash, Card, PayPal, etc. |
| Frequency of Purchases | Weekly, Monthly, etc. |

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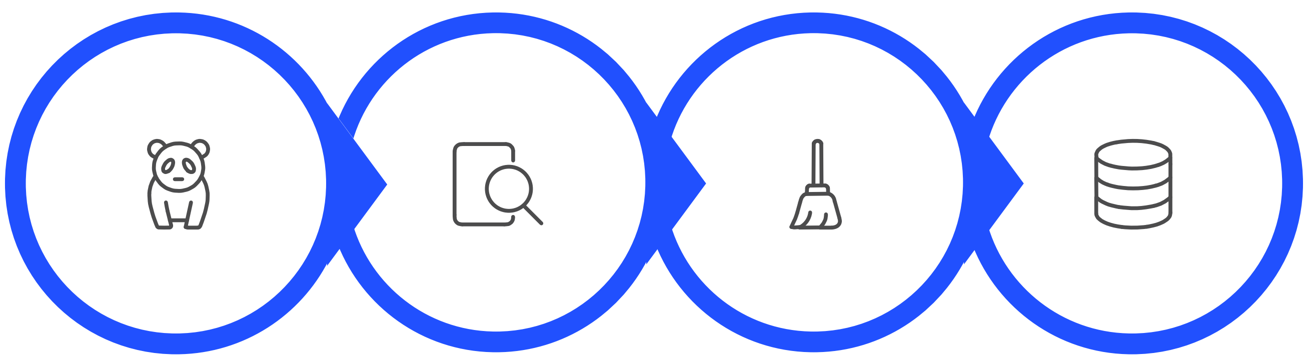
# EDA: Data Cleaning in Python

Load

Explore

Clean

Export

### Key Cleaning Decisions

#### Category Correction

Mapped items to correct categories via dictionary

#### Missing Size

Electronics/Accessories → "Not Applicable"; Clothing → mode

#### Review Rating

Filled with product-level mean (advanced imputation)

#### Previous Purchases

Missing = 0 (no prior purchase)

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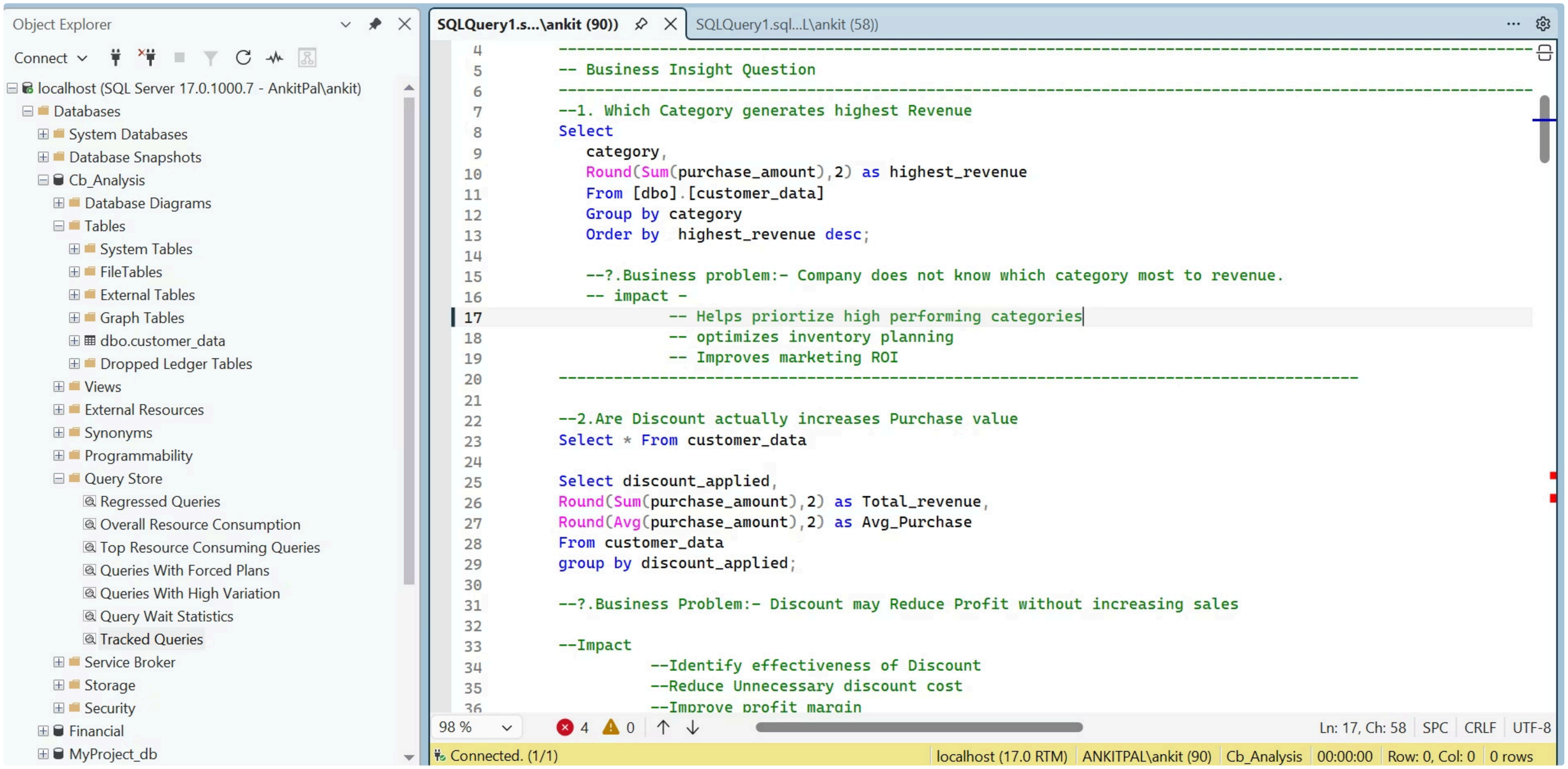
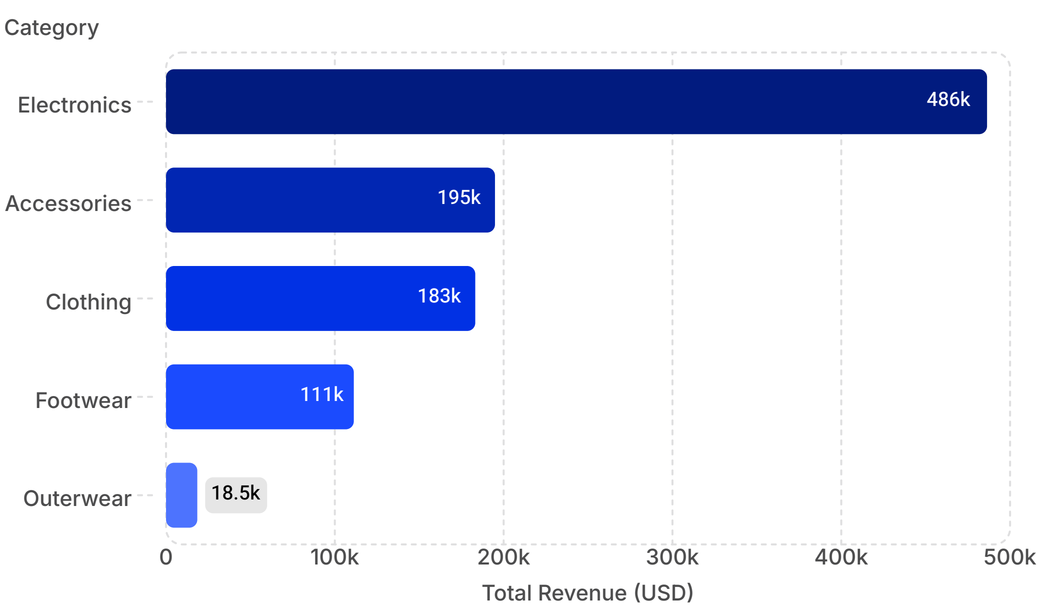
# SQL Analysis: Revenue Insights




## Key Findings

### Electronics #1

$486K — top revenue category by far

### Discounts Work

Avg spend $219 with discount vs. $182 without

### Male Leads

$391K vs. Female $352K vs. Other $251K

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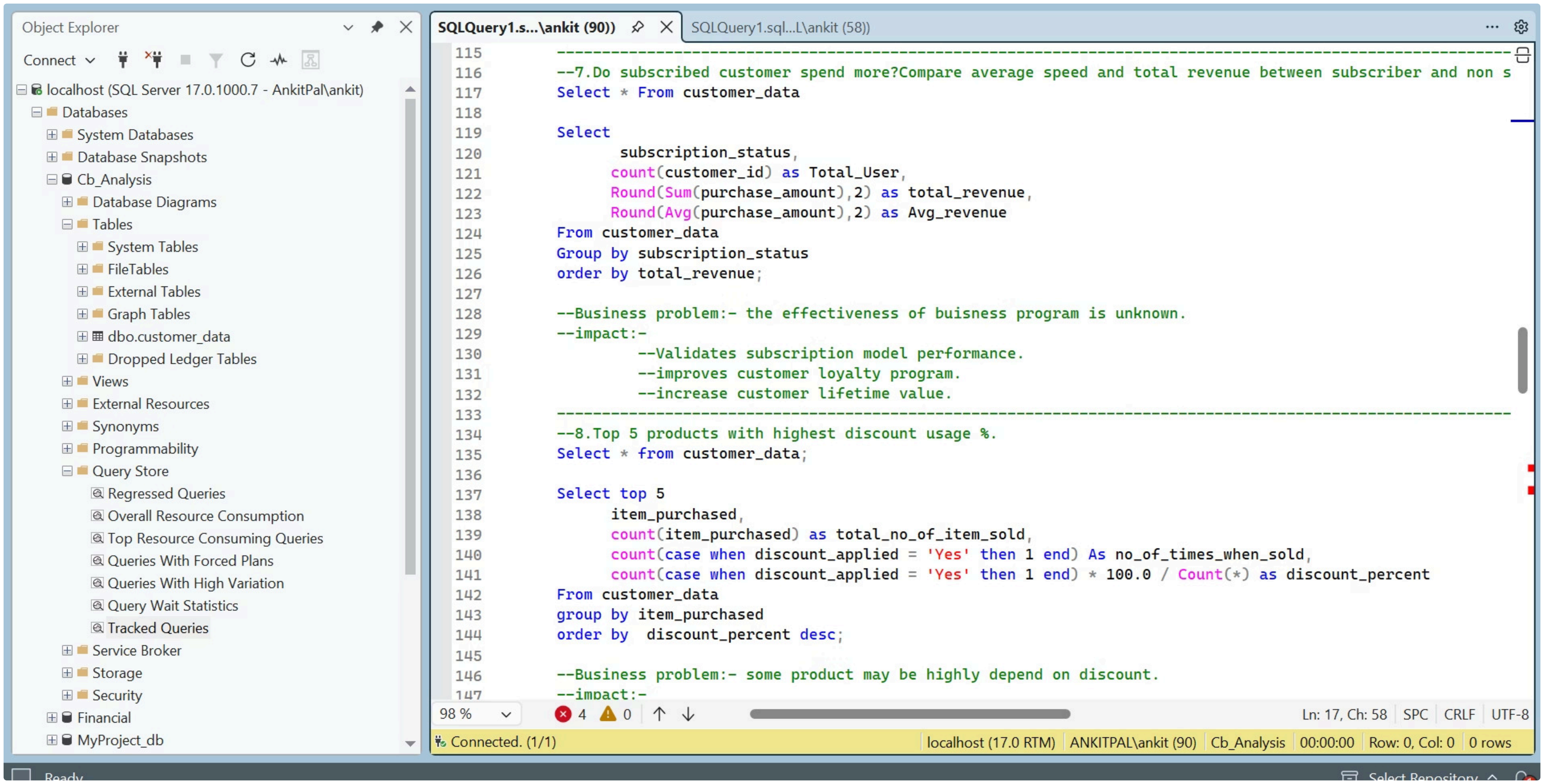
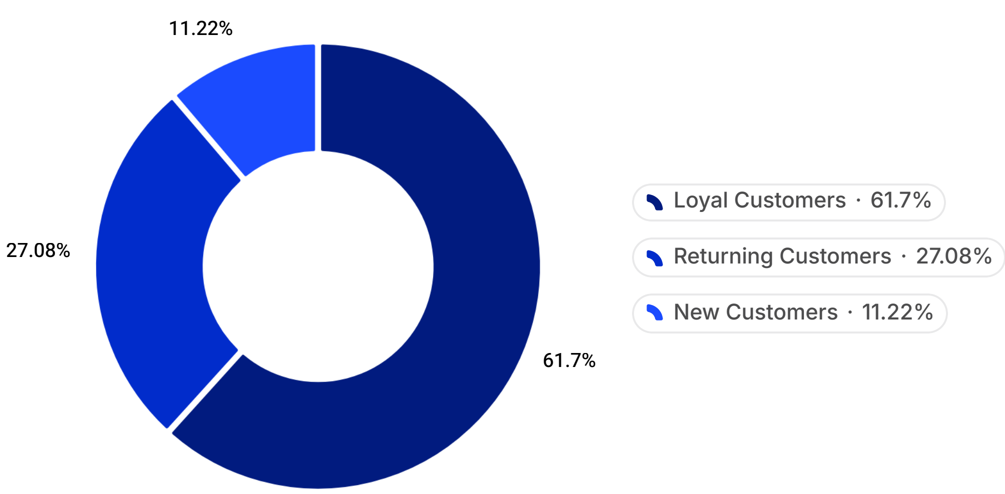
# SQL Analysis: Customer Segments & Subscriptions




## Subscription Impact

| Status | Customers | Avg Spend |
| --- | --- | --- |
| Subscribed | 1,579 | $270.58 |
| Not Subscribed | 3,421 | $165.72 |

Subscribers spend **63% more** on average than non-subscribers.

## 51+ Age Group

Contributes **40.2%** of total revenue ($399K) — the dominant segment.

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# SQL Analysis: Ratings & Top Products

```sql
----------------------------------------------------------------------------------------------------------
--5.Which are top/bottom 5 products with the highest average review rating?

Select * From customer_data

Select Top 5
       item_purchased,
       Round(Avg(review_rating),2) As avg_rating
From customer_data
Group by item_purchased
Order by avg_rating Desc;

Select Top 5
       item_purchased,
       Round(Avg(review_rating),2) As avg_rating
From customer_data
Group by item_purchased
Order by avg_rating Asc;

--Business Problem:- No visibility into product performance based on customer satisfaction?
```

### Top 5 by Rating

| Item | Avg Rating |
|---|---|
| Gloves | 3.86 |
| Sandals | 3.84 |
| Boots | 3.82 |
| Hat | 3.80 |
| Skirt | 3.79 |

### Bottom 5 by Rating

| Item | Avg Rating |
|---|---|
| Phone | 2.94 |
| Headphones | 3.08 |
| Watch | 3.10 |
| Bag | 3.17 |
| Laptop | 3.19 |

### Highest Discount Usage %

| Item | Discount % |
|---|---|
| Laptop | 51.7% |
| Phone | 51.5% |
| Headphones | 51.2% |
| Hat | 50.0% |
| Watch | 49.7% |

⚠ Electronics dominate discount usage — yet also have the **lowest ratings**.

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# Key Takeaways

### Electronics Dominates

$486K revenue — #1 category, but lowest customer ratings

### Loyalty Pays Off

Subscribers spend 63% more; 3,085 loyal customers identified

### Discounts Drive Value

Discounted purchases average $219 vs. $182 without

### 51+ Segment Rules

40.2% of total revenue — the highest-value age group

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